

Solution Blueprint: Aerospace Engineering Order Compliance Review Agent (Simplified Sample)

Automated Airworthiness Audit for Commercial Aircraft Fleet Support

Hybrid Cognitive Architecture Combining Deterministic Engines & Multimodal LLMs on Google Cloud

SIMPLIFIED ARCHITECTURAL SAMPLE & REFERENCE PATTERN

This blueprint is a simplified, synthetic architectural sample created to illustrate a hybrid cognitive pattern combining deterministic rule validation with Google Cloud multimodal vision models (Gemini 3.7 Flash). All document identifiers, engineering order forms, serial numbers, and checklist items are generic, synthetic examples created for reference purposes and do not represent any proprietary customer IP or production data.

Executive Document Information

Attribute	Specification
Client / Domain:	Commercial Aircraft Manufacturing & Engineering Fleet Support
Solution Category:	Airworthiness Compliance Automation (Repair Engineering Orders – REOs)
Technology Stack:	Google Agent Development Kit (ADK) 2.0, Gemini 3.7 Flash Multimodal, Vertex AI Agent Engine, Google Cloud Storage, Gemini Enterprise
Hosting Region:	Google Cloud <code>europa-west1</code> (Belgium) / Runtime: Vertex AI Reasoning Engine
Document Classification:	Enterprise Solution Blueprint & Validated Reference Architecture (Sample)
Status:	Generic Proof-of-Concept & Reference Architecture Pattern
Version:	1.0.0 (Sanitized & Simplified Sample Distribution)

Executive Summary & Solution Highlights

Google Cloud Architecture Statement:

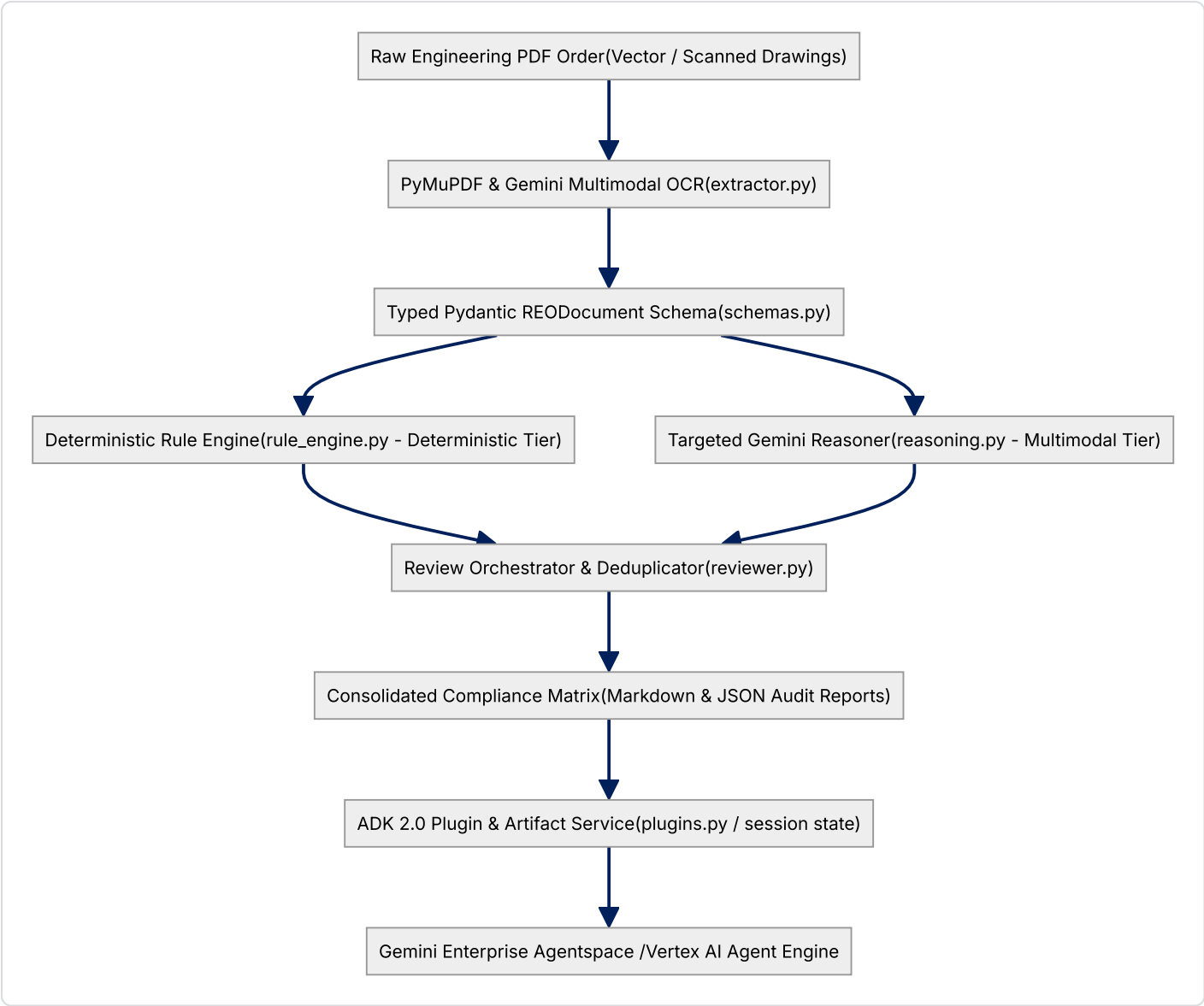
The **Aerospace REO Compliance Review Agent** is a reference architecture designed to automate airworthiness compliance verification for Repair Engineering Orders (REOs). In commercial aircraft fleet support, engineering orders authorize structural repairs. Reviewing these multi-page documents against statutory airworthiness checklists is safety-critical and labor-intensive.

This solution demonstrates a **hybrid cognitive architecture**:

- High-Speed Deterministic Validation (Deterministic Tier):** Fast regex and structural validation for formal parameters (document numbering, serial number ranges, sheet pagination, revision consistency) running in under 50 milliseconds.
- Targeted Multimodal LLM Reasoning (Multimodal Vision Tier):** Gemini 3.7 Flash multimodal vision for spatial diagram orientation (`FWD/AFT/INBD/OUTBD`), grid scale callouts, and mandatory imperative grammar.
- ADK 2.0 Enterprise Governance:** Lifecycle interceptors (`ReoComplianceAuditPlugin`), session-scoped immutable artifact persistence, and continuous evaluation datasets (`.evalset.json`).

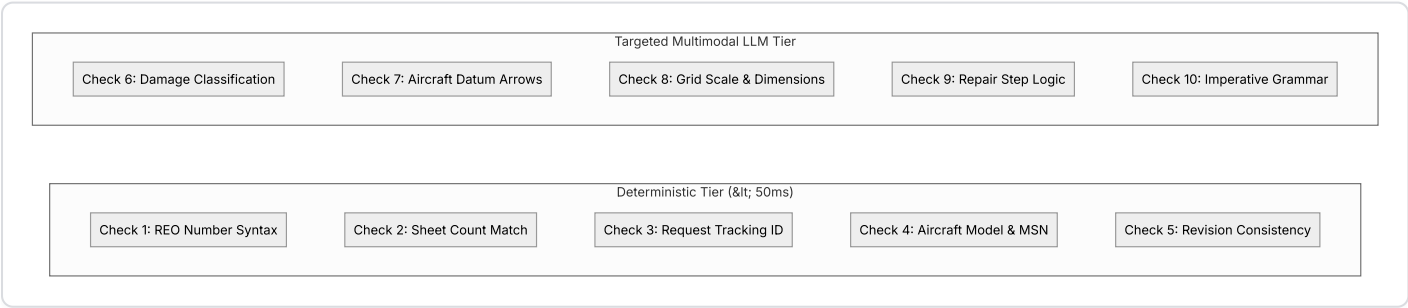
 Sub-Second Deterministic Rules	 Multimodal Visual Precision	 Regulatory Auditability	 Agentspace Integration
< 50 ms Latency for deterministic checks	Gemini 3.7 Flash Vision Datum & grid inspection	ADK 2.0 Interceptors Immutable session state	Gemini Enterprise & CLI Automated approval flow

1. End-to-End Hybrid Cognitive Architecture



2. Representative 10-Point Airworthiness Checklist

The agent partitions the compliance checklist into two complementary execution tiers:



Detailed Compliance Matrix Mapping

#	Checklist Rule	Category	Execution Engine	Verification Logic
1	REO # Format & Header Consistency	Header Integrity	Deterministic	Regex <code>^RE0-\d{2}-\d{2}-\d{4,5}\$</code> validated across cover and headers.
2	Sheet Count vs Page Total	Pagination	Deterministic	Matches <code>1 OF N</code> declaration against physical extracted page count.
3	Technical Request Identifier	Tracking Authorization	Deterministic	Validates tracking identifier syntax (e.g., <code>REQ-100482</code>) or approved standard format.
4	Aircraft Model & Serial Applicability	Fleet Effectivity	Deterministic	Verifies model code (<code>AERO-JET-300</code>) and serial range (<code>MSN 10001-10999</code>).
5	Revision Consistency & Reissue Statement	Version Control	Deterministic	Verifies revision headers (<code>REV --</code> , <code>REV -A</code>) and reissue notice syntax.
6	Damage Characterization & Mode	Structural Inspection	LLM Reasoner	Validates damage type (corrosion, crack, dent) and residual thickness callouts.
7	Drawing View & Datum Orientation	Visual Validation	LLM Reasoner	Confirms aircraft coordinate arrows (<code>FWD</code> , <code>AFT</code> , <code>INBD</code> , <code>OUTBD</code>) on diagrams.
8	Grid Scale & Dimension Callouts	Visual Validation	LLM Reasoner	Validates presence of explicit scale references (e.g., <code>10x10 mm grid</code>) and measurement units.
9	Repair Workflow Sequence	Procedure Logic	LLM Reasoner	Verifies logical sequence: cleanup → NDT inspection → fastener installation → coating.
10	Imperative Verb Grammar	Compliance Syntax	LLM Reasoner	Enforces active imperative instruction phrasing (<code>REMOVE</code> , <code>INSPECT</code> , <code>INSTALL</code> , <code>APPLY</code>).

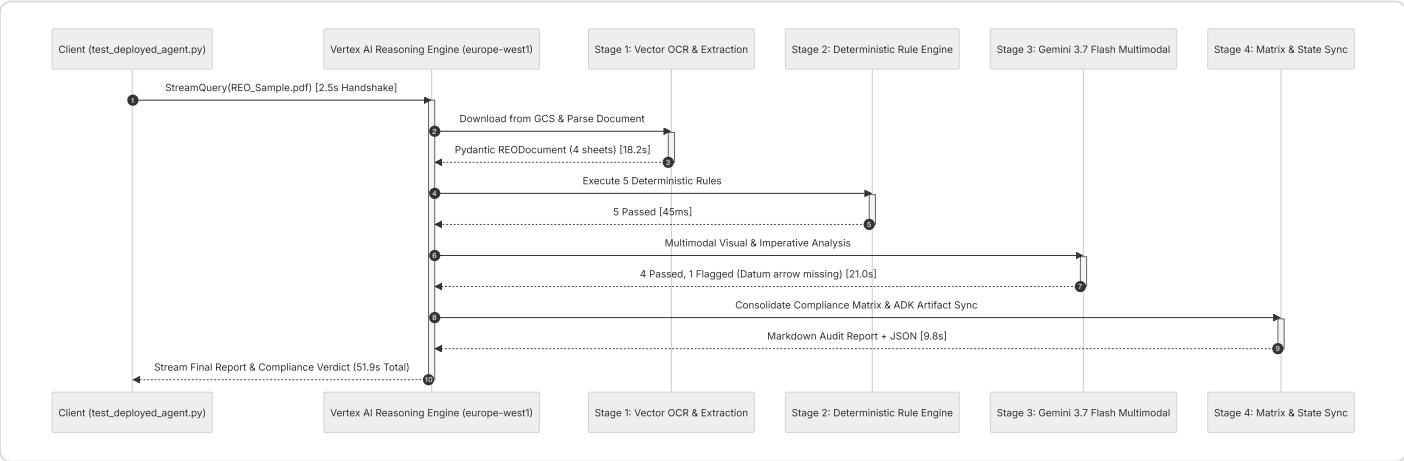
3. Simulated Execution Telemetry & Performance Trace

The agent architecture was benchmarked on Google Cloud Vertex AI Reasoning Engine via gRPC streaming telemetry:

SAMPLE AUDIT EVALUATION SUMMARY (SYNTHETIC BENCHMARK)								
Sample Document	REO Reference	Rev	Pgs	Pass	Fail	Warn	Verdict	
REO-DEMO-53-WING-REPAIR-REV-A.pdf	REO-53-01-1001	-A	6	9	1	0	⚠️	REVIEW
REO-DEMO-54-PYLON-BRACKET-REV--.pdf	REO-54-02-2002	--	4	10	0	0	✅	PASS
REO-DEMO-57-EMPENNAGE-STIFFENER-REV-B.pdf	REO-57-03-3003	-B	8	8	2	0	❌	FAIL

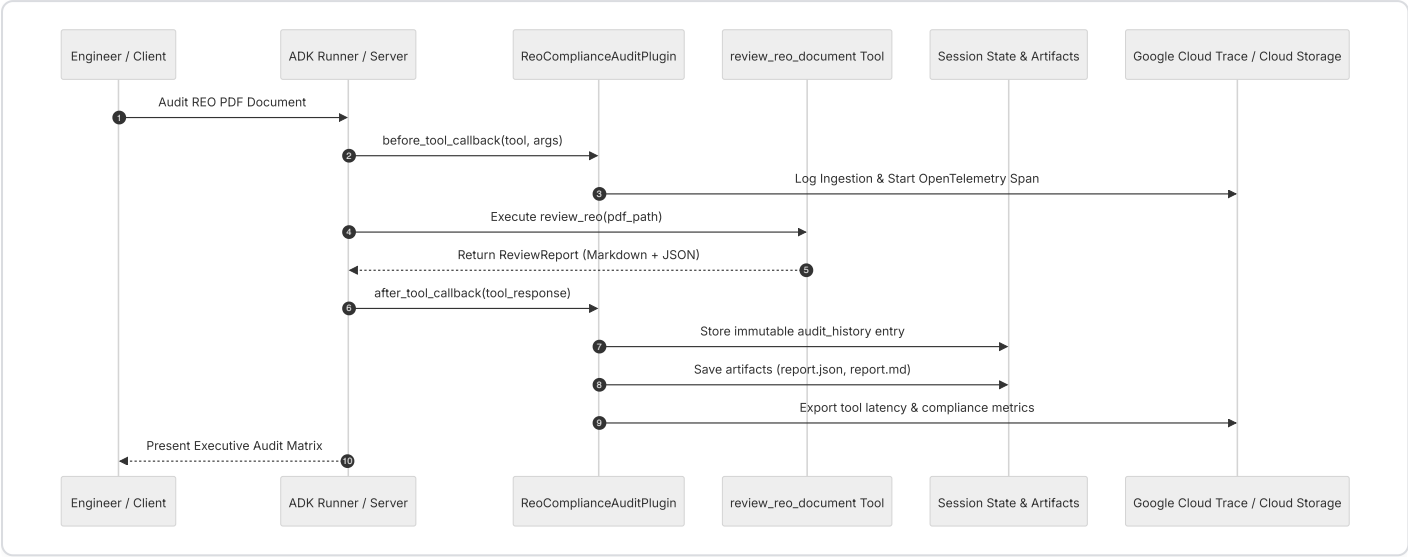
Latency Breakdown by Phase (Sample 4-Page Engineering Order)

Phase	Pipeline Component	Latency	% of Total	Operational Description
Phase 1	Session Handshake & Connection	2.5s	4.8%	Client TLS handshake and session state initialization
Phase 2	Stage 1/4 - Vector OCR & Page Extraction	18.2s	35.0%	Storage download, rasterization, and vector box parsing
Phase 3	Stage 2/4 - Deterministic Rule Engine	45ms	0.1%	High-speed regex, structural validation, and sheet math execution
Phase 4	Stage 3/4 - Targeted Multimodal LLM Reasoning	21.0s	40.4%	Gemini 3.7 Flash visual orientation and imperative grammar checks
Phase 5	Stage 4/4 - Compliance Matrix & State Sync	9.8s	18.9%	Markdown matrix compilation, JSON serialization, and artifact persistence
Phase 6	Stream Finalization & Client Delivery	0.4s	0.8%	Stream flush and client UI formatting
Total	End-to-End Automated Audit	51.9s	100%	Full engineering compliance audit completed



4. ADK 2.0 Governance & Audit Architecture

To meet regulatory auditability requirements, the agent incorporates ADK 2.0 enterprise governance features:



Key Governance Invariants

- Lifecycle Interceptors** (`ReoComplianceAuditPlugin`): Intercepts tool execution to record engineer identity, file digests, and execution timings. Automatically persists findings into `context.state["audit_history"]`.
- Session-Scoped Artifact Persistence**: All intermediate outputs (bounding boxes, extracted text, finding summaries) are archived to Cloud Storage (`gs://[STAGING_BUCKET]/artifacts/{session_id}`).
- CI/CD Regression Sets**: Evaluation datasets (`eval_set.evalset.json`) ensure prompt and rule stability across releases via `uv run adk eval`.

5. Deployment & Production Operations

Cloud Runtime Configuration

- Platform**: Google Cloud Vertex AI Agent Engine (Reasoning Engine)
- Container Region**: `europe-west1` (Low-latency EU sovereign runtime)
- Model Endpoint**: `gemini-3.7-flash` (Vertex AI Global / Regional Endpoint)
- Agentspace Integration**: Discovery Engine Agentspace Assistant registration

```
# Verify local compliance test suite
uv run python scripts/test_samples.py

# Deploy to Vertex AI Agent Engine
uv run python scripts/deploy_agent.py --region europe-west1

# Register agent with Gemini Enterprise Agentspace
bash scripts/register_agent.sh --app-id [AGENTSPACE_APP_ID]
```